

Kunal tyagi

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SKILLS

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- **Languages:** Python, C
 - **Frameworks:** Langchain, Langgraph, FastApi
 - **Databases:** PostgreSQL, SQL
 - **Tools:** Git, GitHub, Linux, VS Code
 - **Core Concepts:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Fast APIs
 - **Soft Skills:** Analytical Thinking, Collaboration, Persistence, Innovation

PROJECTS

RAG Based Chatbot | [GitHub](#)

Feb 2026

- Developed a Retrieval-Augmented Generation (RAG) system using vector embeddings and similarity search to ground LLM responses in custom documents.
- Built an end-to-end ingestion pipeline to parse, chunk, and index unstructured text into a vector database for semantic retrieval.
- Implemented dynamic context retrieval to minimise hallucinations, optimise token limits, and deliver accurate, source-backed answers.
- Technologies: Python, Streamlit, Langchain, Chromadb

Terminal-Based AI Coding Assistant| [GitHub](#)

Nov 2025

- Engineered a terminal-based AI coding agent with custom tool-use capabilities for automated file I/O, context retrieval, and command execution.
- Optimised LLM performance and cost efficiency by implementing streaming API responses, dynamic context window management, and prompt caching.
- Built an interactive terminal UI with rich syntax highlighting, real-time diff previews.
- Technologies: Python, rich, click.

Adaptive Frequency-Hopping 2-Way Communication System

Mar 2025

- Designed a dual-microcontroller (ESP32 & Arduino Nano) 2-way communication device with dynamic multi-band RF transmission.
- Built an automated frequency-hopping algorithm to detect signal degradation and seamlessly switch channels to maintain connection stability.
- Implemented low-latency bidirectional data sync and cross-chip serial communication protocols for reliable real-time messaging.

CERTIFICATIONS

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| • Database Management System Part - 1 Infosys Certificate | Feb 2026 |
| • Data Analysis with Pandas and Python Infosys Certificate | Mar 2026 |
| • Programming Fundamentals using Python Infosys Certificate | June 2026 |

ACHIEVEMENTS

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| • Solved 150+ Python programming problems on Codetantra, demonstrating consistent programming practice and logical reasoning. | Jan 2025 |
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EDUCATION

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|---|---|
| • Lovely Professional University
Bachelor of Technology - Computer Science and Engineering: CGPA: 8.80 | Punjab, India
Aug 2025 - Present |
| • Indraprastha Public School
Intermediate: Percentage: 80% | Baghpat, Uttar Pradesh
Aug 2022 - Mar 2024 |